

NORMAN BUI

Ottawa, ON | +1 613-983-3217 | normanbui23@gmail.com | linkedin.com/in/nbui23 | github.com/nbui23

EDUCATION

Carleton University | *Bachelor of Computer Science (Honours), 3.9/4.0 GPA* Graduated Spring 2026
• Coursework: Data Structures & Algorithms, Operating Systems, OOP, Programming Paradigms, DBMS, Web Development

EXPERIENCE

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025
• Improved CI/CD load-test reliability from **35% to 100%** by fixing auth failures and unstable workflows
• Integrated **10+** game features into load tests for high-traffic performance coverage
• Reviewed automated game-testing architecture and recommended soak-test reliability improvements

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025
• Built TypeScript/Java validation and repair infrastructure for Fusion manufacturing data, including an **O(n)** repair algorithm that improved throughput **8x**
• Deployed GraphQL validation and repair endpoints with **100%** Mocha, Chai, and Sinon coverage
• Fixed permission and error-handling gaps in sensitive manufacturing data workflows

Kongsberg Geospatial | *Software Developer Intern* May 2024 - Aug 2024
• Built client-facing C++/Qt geospatial demos for aviation, defense, and government use cases
• Integrated LiDAR, WMTS, and S-57 GIS data into real-time mapping and tracking workflows
• Developed automated airspace classification framework, removing manual data categorization

Health Canada | *Data Scientist Intern* Sep 2023 - Dec 2023
• Architected TDD scraping framework with unittest, expanding sources by **113%** with **100%** coverage
• Developed toxicity classifiers with pandas, NumPy, SciPy, and scikit-learn
• Rebuilt chemical search with BS4/lxml error handling, raising query success by **20%**

NAV CANADA | *Software Developer Intern* May 2023 - Aug 2023
• Optimized C++/Qt GIS tool, reducing storage **60%** and improving analysis speed **30%**
• Fixed air traffic control rendering bugs, improving frame rate by **16%**
• Automated network setup with PowerShell, reducing manual configuration by **95%+**
• Improved geospatial data import and validation workflows for internal air-traffic tooling

PROJECTS

Orbital Refueling Simulator | *Python, Streamlit, scikit-learn, pytest* github.com/nbui23/orbital-refueling
• Modeled **410-second** refueling mission across **9** phases, **9** anomaly classes, and deterministic generation
• Separated hard-limit engineering rules from **9** phase-aware IsolationForest drift models
• Implemented perturbation-based attribution to rank telemetry channels driving each phase-specific alert

TwinQuery | *Python, LangGraph, FastAPI, Streamlit, PostgreSQL/PostGIS, Ollama, RAG, Docker* github.com/nbui23/twinquery
• Built open-source digital-twin query assistant enabling non-technical users to ask plain-English questions over building-stock and retrofit datasets
• Implemented safe Text-to-SQL pipeline with deterministic guardrails, fallback query templates, and PostgreSQL/PostGIS execution over Ottawa building-footprint geometry
• Integrated local RAG over retrofit guidance with hybrid synthesis, grounding LLM answers in both database rows and retrieved document sources

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, Kotlin, Java, SQL, HTML/CSS, PHP
Frameworks & Runtimes: Next.js, Node.js, Express.js, Electron, Qt, React Native, Expo, Spring Boot, FastAPI, Flask, Streamlit, .NET, ASP.NET
AI / ML / Data: LLMs, RAG, LangGraph, LangChain, FAISS, Ollama, SBERT, scikit-learn, IsolationForest, pandas, NumPy, matplotlib, PyTorch, ONNX Runtime
Databases & APIs: PostgreSQL/PostGIS, pgvector, MySQL, SQLite, MongoDB, Firebase, Neo4j, Prisma, GraphQL
Tools & Practices: Git, AWS, CI/CD, Docker, Docker Compose, Ubuntu, TDD, pytest, Mocha, Chai, Sinon, unittest, BS4, Selenium, Cucumber, PowerShell, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese